

WEEK 3

SQL & Excel for Data Analytics

Student	Tiwari Aman Dhananjay
Assignment	Week 3 — SQL & Excel
Duration	1 Week (7 Days)
Total Marks	100
Tools	Microsoft SQL Server / OneCompiler / Microsoft Excel

Practical Data Analytics Assignment

1. Assignment Overview

This report presents the Week 3 practical assignment on SQL and Microsoft Excel for Data Analytics. The work covers relational data querying, spreadsheet functions, lookups, summarization and visualization.

The SQL section covers SELECT, WHERE, ORDER BY, aggregate functions, GROUP BY, HAVING, joins and subqueries. The Excel section covers formatting, sorting, filtering, conditional formatting, IF, COUNTIF, SUMIF, VLOOKUP, XLOOKUP, Pivot Table and charts.

Task Breakdown

Task	Topic	Marks
1	Database basics & SELECT	10
2	WHERE, ORDER BY & aggregates	15
3	GROUP BY & HAVING	10
4	SQL Joins	15
5	SQL Subqueries	10
6	Excel formatting, sorting & filtering	10
7	Conditional formatting & functions	15
8	VLOOKUP & XLOOKUP	10
9	Pivot Table & Charts	5

Learning Objectives

- Retrieve and filter structured data using SQL.
- Summarize records using aggregate functions and GROUP BY.
- Connect related tables using SQL joins.
- Use subqueries to answer analytical questions.
- Prepare and analyze tabular data in Excel.
- Apply IF, COUNTIF and SUMIF.
- Use VLOOKUP and XLOOKUP for related-table retrieval.
- Create a summary and chart for reporting.

2. SQL Tasks

Task 1 — Database Basics & SELECT

Creates the Students table and demonstrates selecting all records, selected columns and column aliases.

```
SELECT * FROM Students; SELECT Name, Department, Marks FROM Students; SELECT Name AS Student_Name, Marks AS Final_Marks FROM Students;
```

Task 2 — WHERE, ORDER BY & Aggregate Functions

Filters records, sorts results and calculates COUNT, SUM, AVG, MIN and MAX.

```
SELECT * FROM Students WHERE Marks >= 80; SELECT Name, Department, Marks FROM Students ORDER BY Marks DESC; SELECT AVG(Marks) AS Average_Marks FROM Students;
```

Task 3 — GROUP BY & HAVING

Groups students by department, calculates counts and average marks, then filters groups with HAVING.

```
SELECT Department, COUNT(*) AS Student_Count, AVG(Marks) AS Average_Marks FROM Students GROUP BY Department HAVING AVG(Marks) >= 80;
```

Task 4 — SQL Joins

Uses related tables to demonstrate INNER JOIN, LEFT JOIN and RIGHT JOIN through Department_ID.

```
SELECT S.Student_ID, S.Name, D.Department, D.HOD FROM Students_Department S INNER JOIN Departments D ON S.Department_ID = D.Department_ID;
```

Task 5 — SQL Subqueries

Finds employees earning above average salary and products priced above average using nested SELECT statements.

```
SELECT Name, Salary FROM Employees WHERE Salary > (SELECT AVG(Salary) FROM Employees); SELECT Product, Price FROM Products WHERE Price > (SELECT AVG(Price) FROM Products);
```

3. Excel Tasks

Task 6 — Data Formatting, Sorting & Filtering

The workbook contains a structured sales table with Sale_ID, Product_ID, Product, Region, Quantity, Unit_Price and Sales. The data can be formatted, sorted and filtered for easier analysis.

Task 7 — Conditional Formatting & Functions

The workbook demonstrates IF for sales classification and uses COUNTIF and SUMIF for region-based counting and sales totals.

Task 8 — VLOOKUP & XLOOKUP

The Lookup sheet demonstrates retrieving product information from Product_ID. VLOOKUP uses a table range and column position, while XLOOKUP directly maps lookup and return ranges.

Task 9 — Pivot Table & Charts

The summary section groups sales by region and provides a chart for visual comparison of regional performance.

Workbook Areas

Sheet	Purpose
Data	Base sales dataset
Functions	IF, COUNTIF and SUMIF examples
Lookup	VLOOKUP and XLOOKUP examples
Pivot_Chart	Regional summary and chart
Formatting_Filtering	Formatted, sorted and filter-ready data
Instructions	Workbook usage notes

4. Evidence & Screenshot Checklist

SQL evidence — take five genuine screenshots from OneCompiler after executing the supplied SQL script.

Screenshot	Required content
Task1_SELECT.png	Task 1 heading and SELECT output
Task2_WHERE_AGG.png	WHERE, ORDER BY and aggregate output
Task3_GROUPBY.png	GROUP BY and HAVING output
Task4_JOINS.png	INNER/LEFT/RIGHT JOIN output
Task5_SUBQUERIES.png	Both subquery outputs

Excel evidence

Task	Screenshot should show
6	Formatted table with sorting/filter controls
7	Conditional formatting and IF/COUNTIF/SUMIF results
8	VLOOKUP and XLOOKUP results
9	Regional summary and chart

Important: screenshots should be taken from the actual working environment. Random or unrelated screenshots should not be used as submission evidence.

5. Project Structure & Submission

```
Internova-Week3-SQL-Excel/ ■■■■ SQL/ ■ ■■■■ task1_select.sql ■ ■■■■ task2_where_order_aggregate.sql ■ ■■■■  
task3_groupby_having.sql ■ ■■■■ task4_joins.sql ■ ■■■■ task5_subqueries.sql ■■■■ Excel/ ■ ■■■■  
Internova_Week3_SQL_Excel.xlsx ■■■■ datasets/ ■■■■ output/ ■■■■ SCREENSHOT/ ■■■■ README.md ■■■■  
SUBMISSION_CHECKLIST ■■■■ Week3_SQL_Excel_Report.pdf
```

Submission Checklist

- SQL files for Tasks 1–5 are included.
- Excel workbook for Tasks 6–9 is included.
- Five SQL screenshots are captured from OneCompiler.
- Excel screenshots are captured for Tasks 6–9.
- Datasets and output files are included.
- README and final PDF report are included.
- Google Drive link is prepared with the required files.
- GitHub repository is updated if required.

6. Conclusion

This assignment provides practical exposure to SQL and Excel as complementary data-analysis tools. SQL is used for structured data retrieval, filtering, aggregation, grouping, joining and subquery-based analysis. Excel extends the workflow with formulas, lookups, formatting, summaries and charts.

By completing Tasks 1–9, the project demonstrates a complete basic analytics workflow: prepare data, retrieve information, apply analytical logic, summarize results and present findings in a readable format.

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